РОСТОВСКИЙ ЦСМ

Federal Agency
on Technical Regulating and Metrology
Federal budget institution
«STATE REGIONAL CENTRE OF STANDARDIZATION, METROLOGY AND TESTING IN ROSTOV REGION»
LABORATORY FOR FOOD, AGRICULTURAL PRODUCTS AND CONSUMER GOODS
(FBI ROSTOV CSM)

№ RA.RU.21ПЛ84

AFFIRM
Head of Laboratory
_____ A.A. Starikov
« 02 » September 2025



# ANALYSIS REPORT №16696-25/1

| | |
|---|---|
| Product sample name*: | BARLEY, seal K0056170 |
| Name and address of the company*: | Bureau Veritas Russia |
| Delivery date: | 20.08.2025 |
| Period of analysis: | 20.08 – 25.08.2025 |
| Transport vehicle name*: | M/V "SORMOVSKIY-3054"" |
| Shipment quantity*: | 3000 mts ± 10% |
| Port terminal: | --- |
| Subcontractor: | --- |
| Shipper*: | LLC "TOP GRAIN", 344006, RUSSIA, ROSTOV-ON-DON, VOROSHILOVSKIY AVE., 26, 3RD FLOOR, OFFICE 2 ON BEHALF OF MEDITERRA AGRO TARIM TICARET A.S. ON BEHALF OF CONCORD GRAIN TRADING FZCO |
| Consignee*: | TO ORDER |
| Port of loading*: | AZOV PORT, RUSSIA |
| Port of discharge*: | ONE OF THE MARMARA SEA PORT OR SAMSUN PORT /TURKEY |
| Additional information*: | RUAGRJ25005832 |

## ANALYSIS RESULTS

**NORMATIVE DOCUMENTS AND MEASUREMENT INSTRUMENTATION**

Methodological Instructive Regulations "Identification and isomer-specific determination of poly-CDD and poly-CDF in meat, bird, fish and their by-products, and other fat-containing products by chromato-mass-spectrometry", 1999

Methodological Instructive Regulations 4.1.1023-01. "Isomer-specific determination of polychlorinated biphenyls (PCBs) in food", 2001.

EC 152/2009

Standard Samples: СОП 0501-03 – 0520-03
СОП 0601-03 – 0606-03
BCR – 614 Solution S8
RPCM-220, WHO PCB Mixture

| Poisonous material | Toxic equivalent factor (TEF) | Found concentration | |
|---|---|---|---|
| | | ng/kg | TEQ, ng/kg |
| 1 | 2 | 3 | 4 |
| **Dibenzo-p-dioxins (PCDDs)** | | | |
| 2,3,7,8-TetraCDD | 1 | < 0,025 | < 0,025 |
| 1,2,3,7,8-PentaCDD | 1 | < 0,025 | < 0,025 |


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| 1 | 2 | 3 | 4 |
|---|---|---|---|
| 1,2,3,4,7,8-HexaCDD | 0,1 | < 0,050 | < 0,005 |
| 1,2,3,6,7,8- HexaCDD | 0,1 | < 0,050 | < 0,005 |
| 1,2,3,7,8,9- HexaCDD | 0,1 | < 0,050 | < 0,005 |
| 1,2,3,4,6,7,8- HeptaCDD | 0,01 | < 0,100 | < 0,001 |
| OctaCDD | 0,0003 | < 0,250 | < 0,000075 |
| **Dibenzofurans (PCDFs)** | | | |
| 2,3,7,8- TetraCDF | 0,1 | < 0,025 | < 0,0025 |
| 1,2,3,7,8- PentaCDF | 0,03 | < 0,025 | < 0,00075 |
| 2,3,4,7,8- PentaCDF | 0,3 | < 0,025 | < 0,0075 |
| 1,2,3,4,7,8- HexaCDF | 0,1 | < 0,050 | < 0,005 |
| 1,2,3,6,7,8- HexaCDF | 0,1 | < 0,050 | < 0,005 |
| 1,2,3,7,8,9- HexaCDF | 0,1 | < 0,050 | < 0,005 |
| 2,3,4,6,7,8- HexaCDF | 0,1 | < 0,050 | < 0,005 |
| 1,2,3,4,6,7,8- HeptaCDF | 0,01 | < 0,100 | < 0,001 |
| 1,2,3,4,7,8,9- HeptaCDF | 0,01 | < 0,100 | < 0,001 |
| OctaCDF | 0,0003 | < 0,250 | < 0,000075 |
| OtherTetrachlordibenzodioxins | - | < 0,025 | - |
| OtherPentachlordibenzodioxins | - | < 0,025 | - |
| OtherHexachlordibenzodioxins | - | < 0,050 | - |
| OtherHeptachlordibenzodioxins | - | < 0,100 | - |
| OtherTetrachlordibenzofurans | - | < 0,025 | - |
| OtherPentachlordibenzofurans | - | < 0,025 | - |
| OtherHexachlordibenzofurans | - | < 0,050 | - |
| OtherHeptachlordibenzofurans | - | < 0,100 | - |
| **Dioxin-like PCBs Non-ortho PCBs** | | | |
| PCB - 77 | 0,0001 | < 1,000 | < 0,0001 |
| PCB - 81 | 0,0003 | < 1,000 | < 0,0003 |
| PCB - 126 | 0,1 | < 1,000 | < 0,1 |
| PCB - 169 | 0,03 | < 2,000 | < 0,06 |
| **Dioxin-like PCBs Mono-ortho PCBs** | | | |
| PCB - 105 | 0,00003 | < 1,000 | < 0,00003 |
| PCB - 114 | 0,00003 | < 1,000 | < 0,00003 |
| PCB – 118 | 0,00003 | < 1,000 | < 0,00003 |
| PCB - 123 | 0,00003 | < 1,000 | < 0,00003 |
| PCB - 156 | 0,00003 | < 2,000 | < 0,00006 |
| PCB - 157 | 0,00003 | < 2,000 | < 0,00006 |
| PCB - 167 | 0,00003 | < 2,000 | < 0,00006 |
| PCB - 189 | 0,00003 | < 4,000 | < 0,00012 |
| **Threshold of detectability 2,3,7,8- TetraCDD -$^{13}$C, ng/kg: 0,05** | **Total concentration in TEQ** | | **< 0,259** |
| **non Dioxin-like PCBs** | | | |
| PCB – 28 | --- | < 1,000 | --- |
| PCB – 52 | --- | < 1,000 | --- |
| PCB – 101 | --- | < 1,000 | --- |
| PCB - 138 | --- | < 1,000 | --- |
| PCB – 153 | --- | < 1,000 | --- |
| PCB - 180 | --- | < 1,000 | --- |

ng WHO-PCDD/F- TEQ/kg (88% d.w.)=0.098. Maximum level 0.75 ng WHO-TEQ/kg

ng WHO-PCDD/F/PCBs- TEQ/kg (88% d.w.)=0.259. Maximum level 1.25 ng WHO-TEQ/kg

Dioxin and Dioxin-like PCB's <1.25 ng WHO-PCDD/F-PCB-TEQ/kg in compliance with EU Regulation 277/2012

The sum of non- dioxin like PCB's of the above mentioned sample is below the maximum level of 10 µg/kg relative to a feed with a moisture content of 12 %

Remarks: «-» in «Concentration» column means that the actual value does not exceed the detection limit of the method

Tables of TEFs for risk assesment based on EU Regulations 277/2012

| | | |
|---|---|---|
| WHO (PCDD/PCDF) incl. 0,5*LOD, ng/kg TEQ | < | 0,05 |
| WHO (PCDD/PCDF) incl. 1,0*LOD, ng/kg TEQ | < | 0,10 |
| WHO (Dl PCB) incl. 0,5*LOD, ng/kg TEQ | < | 0,08 |
| WHO (Dl PCB) incl. 1,0*LOD, ng/kg TEQ | < | 0,16 |
| WHO (PCDD/PCDF + Dl PCB) incl. 0,5*LOD, ng/kg TEQ | < | 0,13 |
| WHO (PCDD/PCDF +Dl PCB) incl. 1,0*LOD, ng/kg TEQ | < | 0,26 |

**LABORATORY RESULTS**

based on the analytical results, we confirm that Total concentration in TEQ of PCDD/F + Dioxin-like PCB not exceed the levels indicated in EC/2002/32, EC/2023/915, EU Regulation 277/2012 and its Ammendments

Responsible for the preparation and execution of the test report:

| Engineer | | A. A. Ishchenko |
|---|---|---|
| (Job position) | (Signature) | (Surname) |

End of analysis report